



Cambridge Lower Secondary Checkpoint

SCIENCE

0893/01

Paper 1

October 2025

FINAL MARK SCHEME

Maximum Mark: 50

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Markers were instructed to award marks. It does not indicate the details of the discussions that took place at a Markers' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the End of Series Report. Cambridge will not enter into discussions about these mark schemes.

This document has **12** pages.

General guidelines on marking

Many descriptive answers can be expressed in a variety of ways. Professional judgement can be used in these cases, providing it matches the marking points and further information in the mark scheme.

Answers may have words spelt incorrectly. Credit is normally given for phonetically correct answers, unless the word has a scientifically different meaning. For example, where the answer should be antennae, credit will be given for antenna but not for anthen (too close to anther).

Only the science is being assessed so answers do not need to be grammatically correct. Significant figures will be indicated in the question or in the mark scheme. Unless specified all marking points are independent.

Annotations and abbreviations

/ or	alternate responses for the same marking point
() brackets	the words or units in brackets do not need to be stated, for example, (recycles or releases or provides) minerals = minerals scores the mark
<u>Underline</u>	exact word is required
Accept	an acceptable response
Do not accept	indicates an incorrect response that would contradict another otherwise correct alternative
Ignore	indicates an irrelevant answer that is not creditworthy. Full marks can still be achieved even with answers that are ignored.
Note	provides extra information when necessary
ecf	error carried forward; marks are awarded if an incorrect response has been carried forward from earlier working, provided the subsequent working is correct
ora	or reverse argument; for example, as mass increases, volume increases could be written as mass decreases, volume decreases

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Question	Answer	Marks	Further Information
1(a)	chromosomes	1	Accept sex chromosome
1(b)	gene(s)	1	Accept allele Ignore recessive or genetic
1(c)	yes ticked (no mark) there is a Y chromosome	1	if no ticked = 0 marks Accept (sex chromosomes are) XY Accept (idea that) if it was a woman sex chromosomes would be XX

Question	Answer	Marks	Further Information
2(a)	E	1	Accept boron or 2.3
2(b)	D	1	Accept potassium or K or 2.8.8.1
2(c)	F	1	Accept helium or He or 2
2(d)	A and D	1	Accept C as an alternative to either A or D Note for A accept Na or sodium or 2.8.1 Note for D accept K or potassium or 2.8.8.1 Note for C accept H or hydrogen or 1

Question	Answer	Marks	Further Information
3	vibrate collide hot(ter) or warm(er) and cold(er) or cool(er)	3	each correct sentence = 1 mark Accept left end to right end Accept heated end to other end

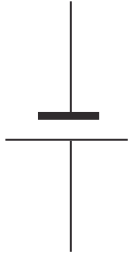
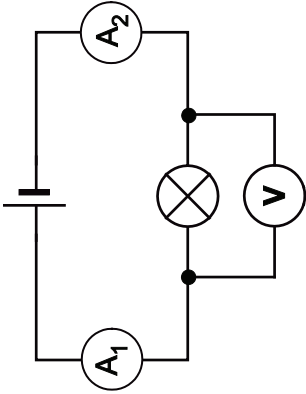
Question	Answer	Marks	Further Information
4	convection current (in the cooking oil) or hot cooking oil moves upwards and cool cooking oil moves downwards movement of cooking oil pushes wood apart or drags the wood apart	2	each correct answer = 1 mark Accept marks from a labelled diagram

Question	Answer	Marks	Further Information
5(a)	M1 (idea that) changes the distance between the lamp and the pondweed M2 counts the number of bubbles produced M3 in a fixed time	3	each correct answer = 1 mark Accept (idea of) measure the time (it takes for the results or collect the results) Accept time to produce a fixed number of bubbles = 2 mark
5(b)	any one from temperature amount of carbon dioxide species of pondweed or size of pondweed	1	Accept use (same) lamp or (same) brightness of lamp Do not accept light intensity or distance Accept concentration of carbon dioxide Accept mass of pondweed or type of plant or (same) pondweed Accept (same) time unless time is changed in (a)

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Question	Answer	Marks	Further Information
5(c)	as light intensity increases the rate of photosynthesis will increase / ora	1	Accept when the lamp is closer photosynthesis is faster / ora
5(d)	(idea that) allows anomalous results to be spotted makes results more reliable	2	each correct answer = 1 mark Accept (idea that) allows anomalous results to be excluded for the mean or do not include anomalous results in the average = 2 marks Accept to get more precise results Do not accept makes results more accurate or fair test
Question	Answer	Marks	Further Information
6(a)	copper + sulfuric acid $\longrightarrow$ copper + carbon dioxide + water carbonate sulfate dioxide	2	correct reactants any order = 1 mark correct products any order = 1 mark Accept = for $\longrightarrow$ Accept correct formulae but words take precedence Do not accept 'and' for a plus sign
6(b)	filtration or by filtering it	1	Accept use filter paper Ignore decanting or sieving

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Question	Answer	Marks	Further Information
6(c)	M1 evaporate off some of the water M2 leave solution to crystallise slowly	2	each correct answer = 1 mark Accept heat to make a saturated solution or heat to reduce the volume by a half or heat until first crystals appear Accept leave solution in a warm place Accept leave warm solution to cool down
Question	Answer	Marks	Further Information
7(a)		1	Do not accept symbol for two or more cells
7(b)	0.55 (A)	1	
7(c)		2	correct circuit symbol (drawn on diagram) = 1 mark Do not accept a line going through the voltmeter circuit symbol unless the voltmeter is added in series correct position of voltmeter in the circuit = 1 mark Ignore missing junction of conductors
7(d)	(ammeter reading) decreases	1	Accept gets smaller or less than 0.55 (A) or numerical answers less than 0.55 (A) Do not accept slower

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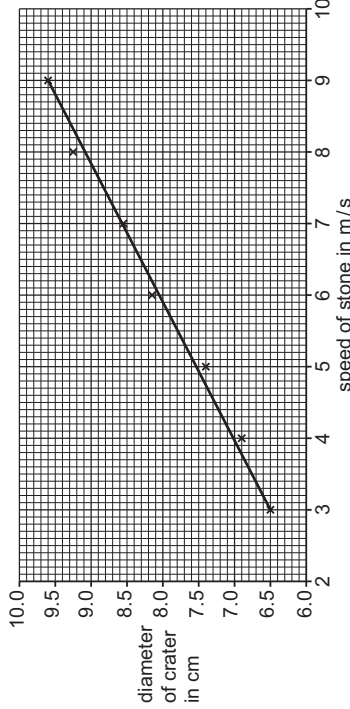
Question	Answer	Marks	Further Information
8(a)	(process A) photosynthesis (process B) respiration or decomposition (process C) combustion	3	each correct answer = 1 mark Do not accept breathing Accept burning
8(b)	any one from sea level change increased global temperature flooding drought (idea of) extreme weather events ice caps melting or glaciers melting or polar ice melting extinction (of animals and plants) or reduce biodiversity (idea of) less food production (idea of) loss of habitat or change of habitat	1	Accept heating up the atmosphere or it will be warmer or increasing temperature or global warming Accept desertification Accept (idea of) extreme weather events, e.g. storms or heatwaves Accept unexpected weather events Note ice melting is not sufficient Note affects the food chain is not sufficient

Question	Answer	Marks	Further Information
9(a)(i)	different sizes (only) one has spots in the dark edge of the wings	2	each correct answer = 1 mark Accept pattern (on wings) or patterns (on butterfly) or white spots on wings Accept other examples of variation that can be seen in the diagrams e.g. shape of body Ignore colour differences
9(a)(ii)	genes or genetics	1	Accept chromosomes or DNA or mutation
9(b)	gametes fuse	2	each correct answer = 1 mark Accept sperm and egg or sex cells Accept join or combine or merge together Accept sperm goes into egg or sperm enters egg = 2 marks Note male cell and female cell is not sufficient

Question	Answer	Marks	Further Information
10(a)	density = mass $\div$ volume or $20 \div 4$ 5 g /cm³	3	5 g/cm³ with or without working = 3 marks 5 with or without working = 2 marks correct equation or evidence of correct substitution into equation = 1 mark Accept density = m/v correct value for density = 1 mark correct units = 1 mark
10(b)	solid B will float it has a density less than water or density less than 1 g/cm³	2	each correct answer = 1 mark Note If solid B sinks = 0 marks for the question Accept ora

Question	Answer	Marks	Further Information
11(a)	(remains the same) amplitude (changes) frequency	2	each correct answer = 1 mark Accept loudness or volume Accept pitch or wavelength
11(b)	(idea that) the pitch increases (idea that) the loudness increases and then decreases	2	each correct answer = 1 mark Ignore references to frequency Accept sound gets softer for loudness decreases Accept volume for loudness Ignore references to amplitude

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Question	Answer	Marks	Further Information
12(a)	point correctly plotted at (4, 6.9)	1	Accept tolerance of $\pm$ half a small square Ignore size of the point
12(b)	acceptable line of best fit judged by eye 	1	Note some points must be above the line of best fit and others below the line to award the mark Accept ecf from (a) Do not accept dot-to-dot
12(c)	(idea of) increases or gets greater (idea of) increases or gets bigger or larger or wider	1	both correct for the mark Accept ora